

Observation of Feshbach resonances between a single barium ion and an ultracold lithium gas

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Albert-Ludwigs-Universität Freiburg



UNI
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21. August 2023

What we do at BaLi

- Co-trapping $^{138}\text{Ba}^+$ and ^6Li .
- Optical trapping of ions
- Looking for interactions in the ultracold regime

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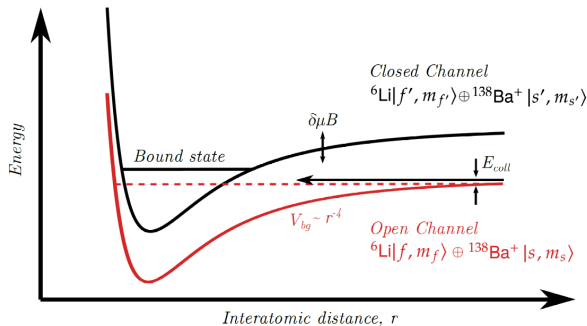
- Co-trapping $^{138}\text{Ba}^+$ and ^6Li .
- Optical trapping of ions
- Looking for interactions in the ultracold regime

Why $^{138}\text{Ba}^+ - ^6\text{Li}$?

- Large mass ratio means lower collision energy
- Combination is chemically stable in their ground states
- Candidates for all-optical trapping

The Feshbach Resonances

An incoming state (open channel) can couple to a bound state (closed channel).
Closer energies mean more mixing between the states.

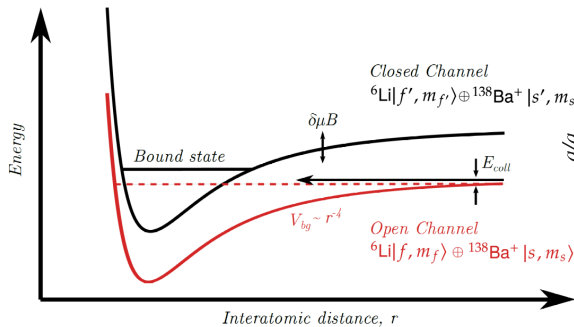


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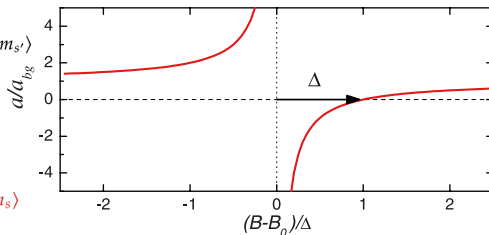
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Closer energies mean more mixing between the states.

Magnetic fields can shift potentials relative to each other (magnetic FR)



$$a = a_{bg} \left(1 - \frac{\Delta}{B - B_0} \right)$$



Chin et al (2010)

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 - strong ($a \rightarrow \pm\infty$)
 - attractive ($a > 0$)
 - repulsive ($a < 0$)
 - switched off ($a = 0$)

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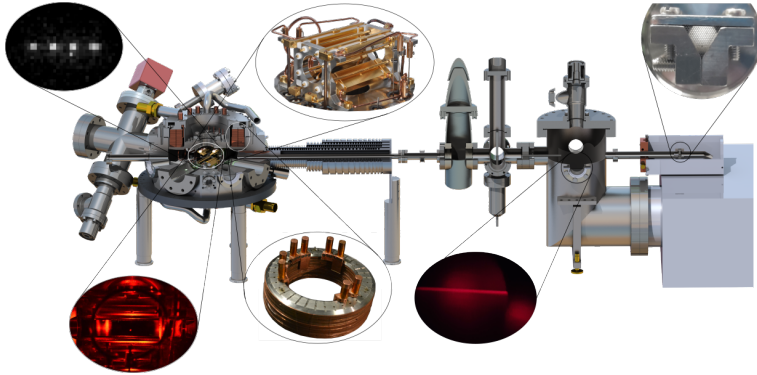
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- Insight into molecular spectra
- Formation of cold molecules in chosen states

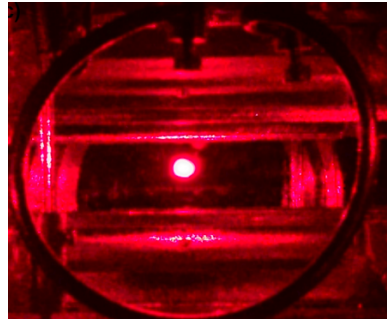
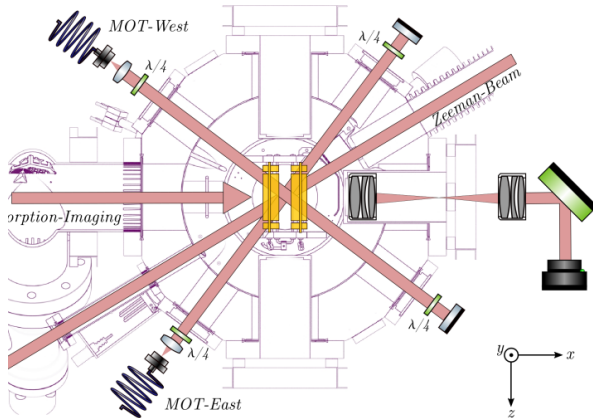
The BaLi Experiment

An apparatus to combine single (to tens) of Doppler cooled Ba^+ -ions with $\approx 2 \cdot 10^4$ ^6Li -atoms at $T \approx 1\mu\text{K}$



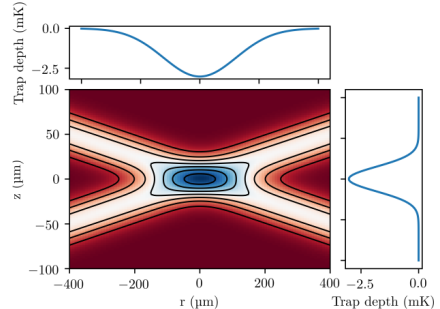
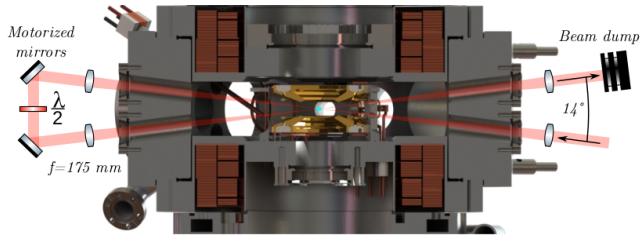
The Atom Trap

Atomic beam cooled with Zeeman slower.
Initially loaded into a MOT of $\sim 10^6$ atoms.



The Atom Trap

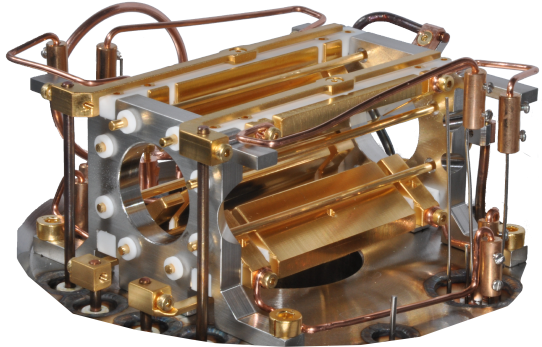
- Atoms transferred from MOT to crossed optical dipole trap (xODT)
- Evaporatively cooled to $T \approx 1 \mu\text{K}$
- Spin polarised
- $\sim 10^4$ ^6Li atoms. Peak density $\sim 10^{13} \text{ cm}^{-3}$.



The Ion Trap

Trap parameters:

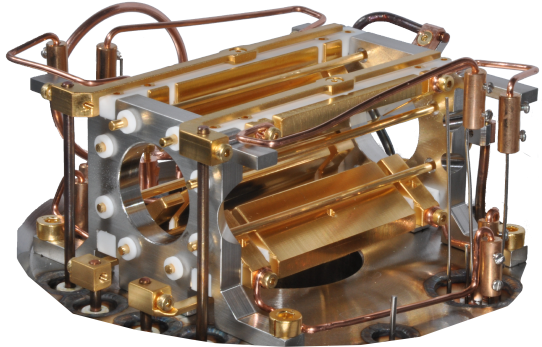
- $\omega_{\text{rad}} \approx 2\pi \times 100 \text{ kHz}$
- $\omega_{\text{ax}} \approx 2\pi \times 10 \text{ kHz}$
- $\Omega_{\text{rf}} \approx 1.4 \text{ MHz}$



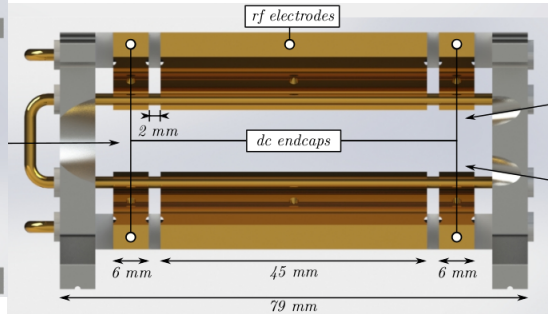
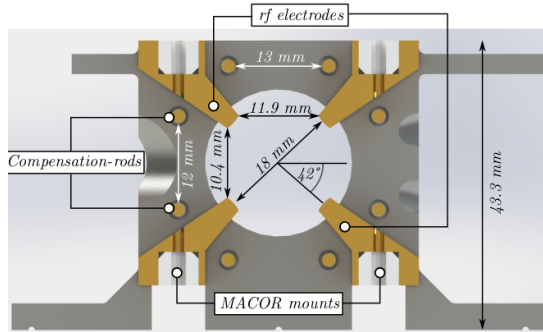
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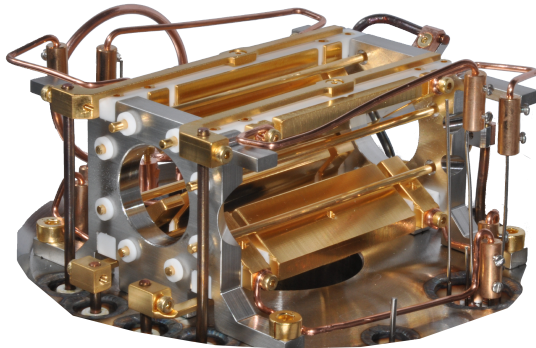


The Ion Trap



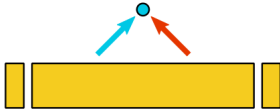
The Ion Trap

- Loaded through ablation loading and photoionisation
- Doppler cooled to near Doppler limit ($T_D = 365 \mu\text{K}$)
- Prepared in ground state spin-mixture
- Detected via fluorescence on CCD
- $\approx 1 \mu\text{m}$ position resolution
- Stray fields compensated to 3 mV m^{-1}



Atom-Ion Interaction Scheme

I.



1. Doppler cool ion

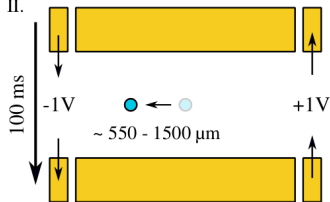
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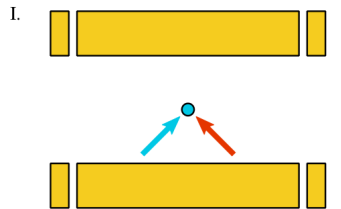
1. Doppler cool ion

II.

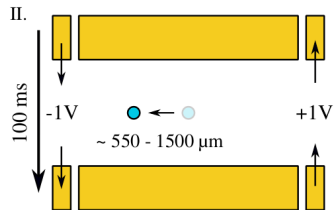


2. Shift ion from centre

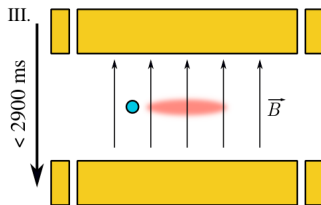
Atom-Ion Interaction Scheme



1. Doppler cool ion

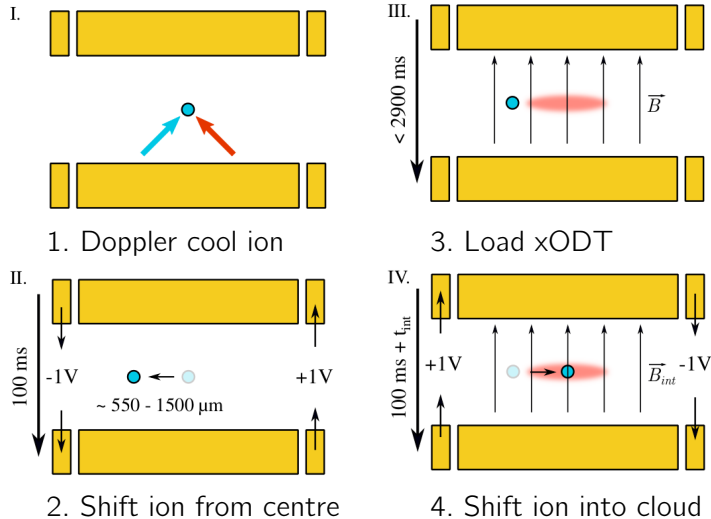


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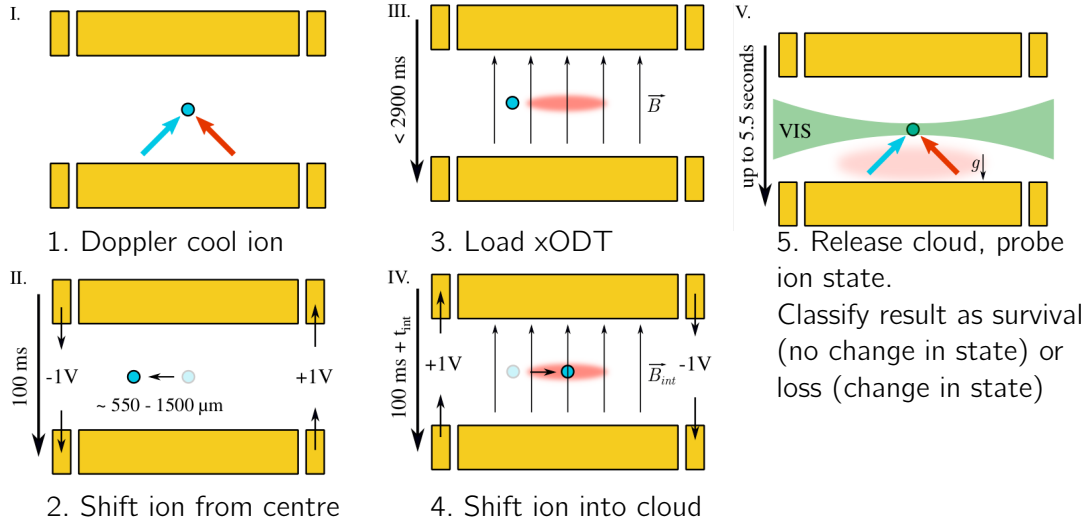


3. Load xODT

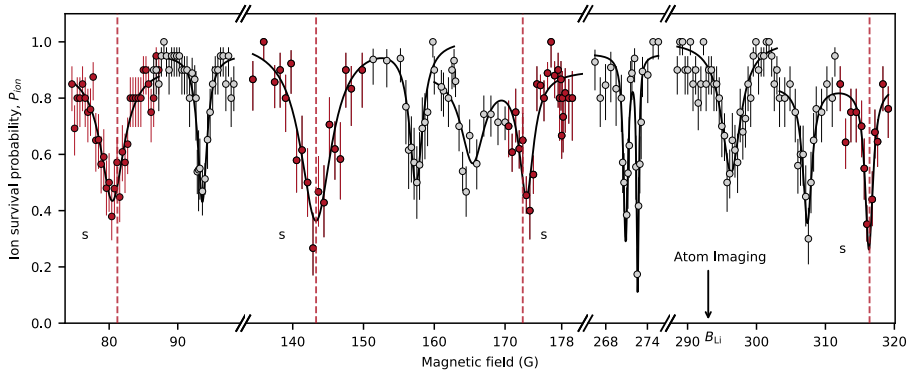
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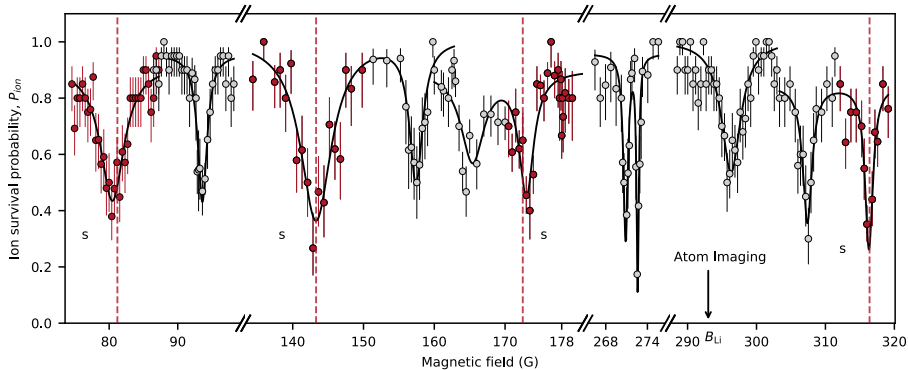
Atom-Ion Interaction Scheme



Feshbach Resonances: Spectrum for Li in $|2\rangle$

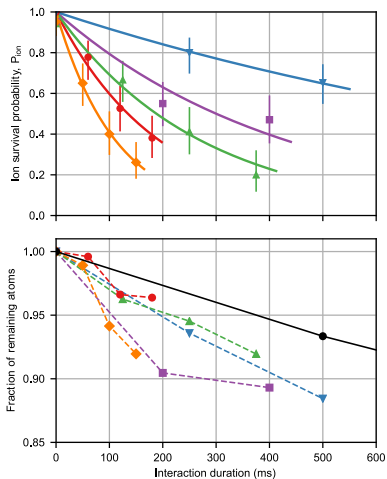


Feshbach Resonances: Spectrum for Li in $|2\rangle$



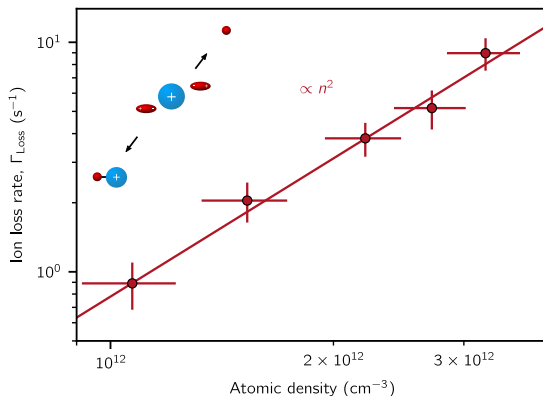
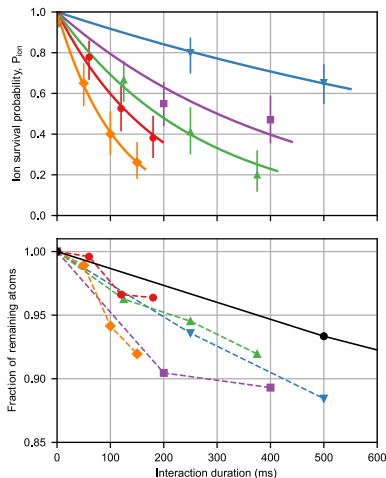
Free fit of singlet and triplet scattering lengths assigns four s -wave resonances

Density Dependence of Ba^+ Loss on Resonance



Vary atomic density and probe ion decay rate.

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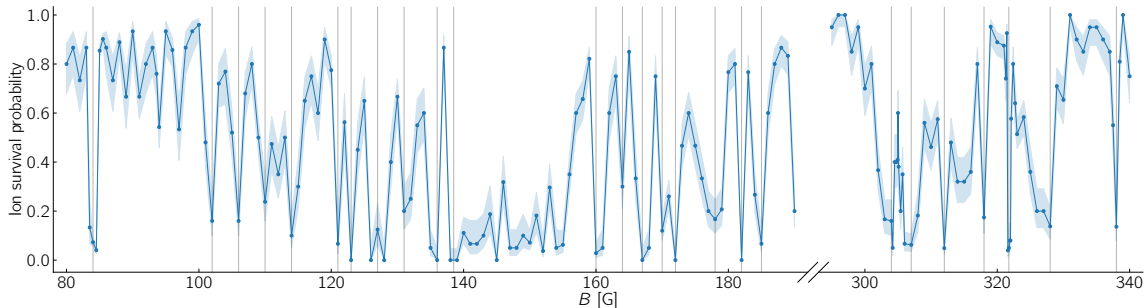


Vary atomic density and probe ion decay rate.
 $\Rightarrow$ TBR is dominant loss process on resonance

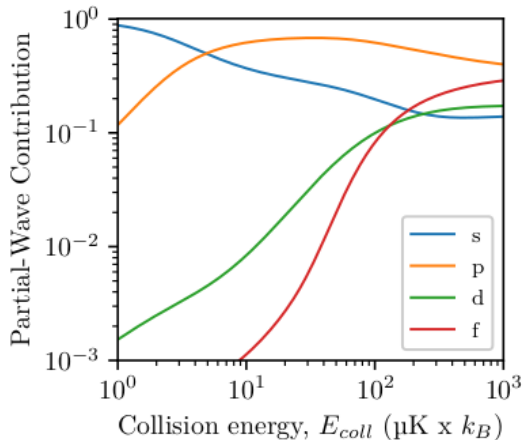
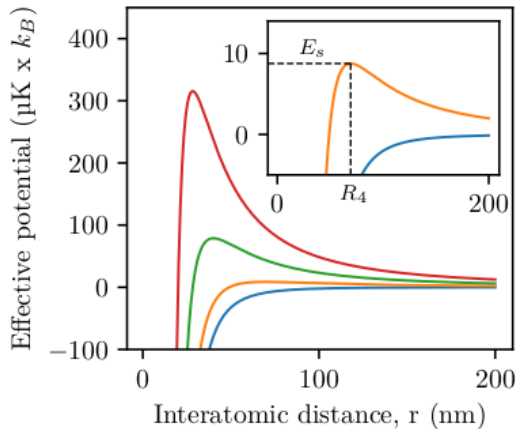
The Forest of Resonances for Li in $|1\rangle$



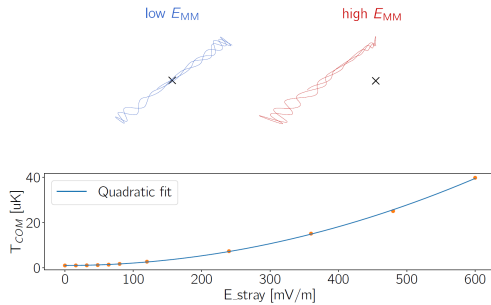
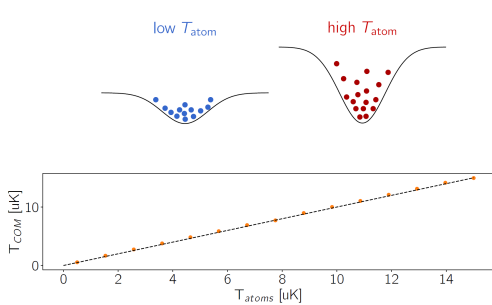
We scanned a range of 160 G and didn't find a region with flat background > 5 G wide.



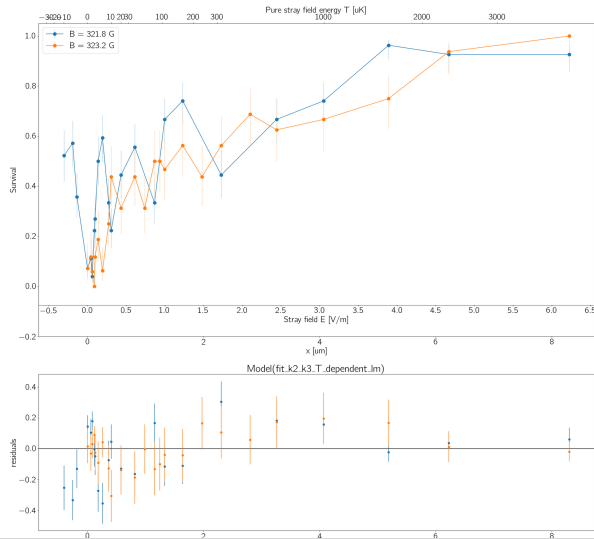
Changing collision energy



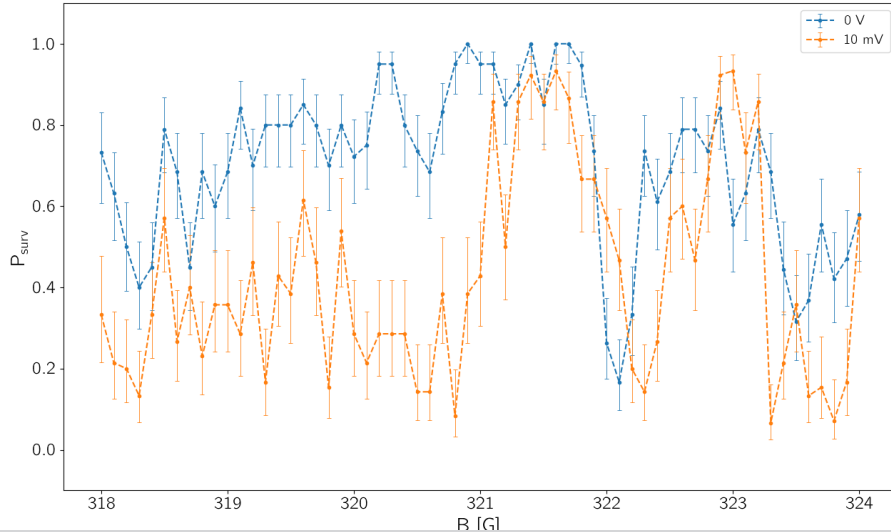
Two ways to tune collision energy



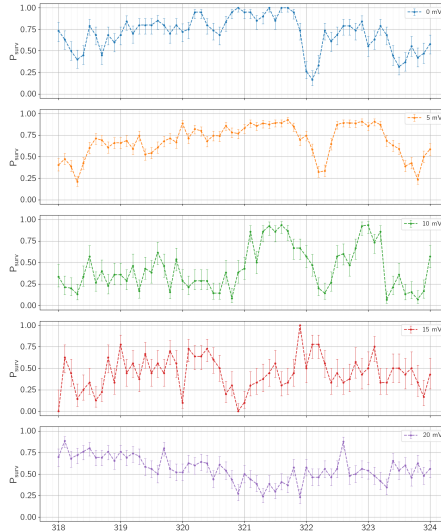
Scanning the offset field



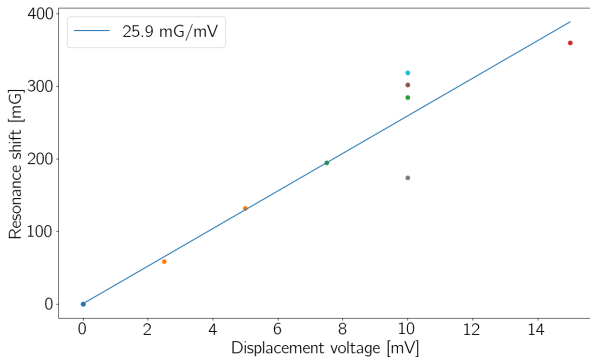
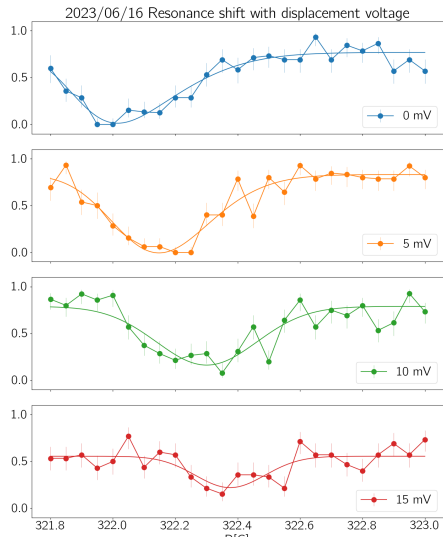
A hotter spectrum



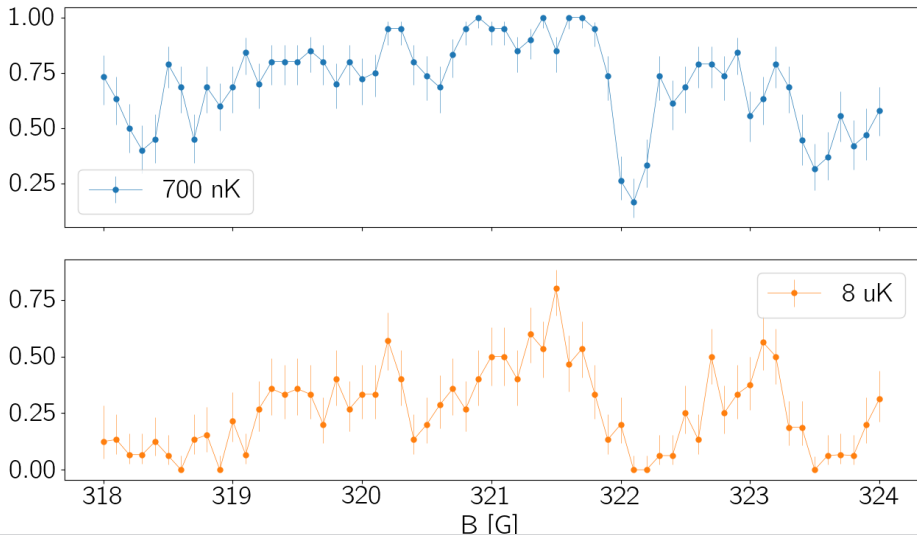
Even hotter spectra



A shifting resonance



Turning up the bath temperature



Near(ish) term:

- Adiabatic ramps for Li
- Spin-polarisation of Ba^+

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Interactions in an all-optical trap

- No micromotion!
- Light-dependent interactions?
- Balancing Ba^+ trap-depth and Li temperature/density

Thank you!



BaLi Team:
Fabian Thielemann, Joachim Welz, Wei
Wu, Pascal Weckesser (former)
Tobias Schätz

Theory collaborators:
Michał Tomza,
Krzysztof Jachymski,
Agata Wojciechowska

